

Assignment 9: Mutation Testing

Objective:

Understand mutation testing through manual and automated approaches.

PHASE 1: MANUAL MUTATION TESTING (2 Hours)

- Write unit tests
- Create at least 20 mutants
- Run tests and identify killed/survived mutants
- Improve test suite
- Achieve >95% mutation score

PHASE 2: AUTOMATED MUTATION PIPELINE (2 Hours)

- Build script to generate mutants automatically
- Execute tests for each mutant
- Track mutation score
- Generate report

Deliverables:

- source_code.py
- test_cases.py
- mutation_pipeline.py
- mutation_report.json
- report.pdf

Focus:

- Edge cases
- Equivalent mutants
- Coverage vs mutation score